

Acute Myeloid Leukemia (AML) - Adult

KEYWORDS / TYPICAL PRESENTATION

- Older adult, acute onset of fatigue, infections, bleeding.
- Blasts on smear with **"Auer Rods"**: Pathognomonic for AML.
- Extreme bruising/bleeding + **abnormal coagulation studies (prolonged PT/PTT, low fibrinogen)**: Suspect DIC from APL (MEDICAL EMERGENCY).
- **Very high WBC (>100k)** + dyspnea, confusion: Suspect Leukostasis (MEDICAL EMERGENCY).

DIAGNOSTIC PATHWAY

- **Best Initial Step**: Peripheral Smear.
- **Confirmatory Test**: Bone Marrow Biopsy (>20% blasts).
 - Critical Molecular Finding: Look for **t(15;17) or PML-RARA** which defines Acute Promyelocytic Leukemia (APL).

Core Principles for AML Management:

1. **If APL is suspected: START ATRA IMMEDIATELY** before genetics.
2. For non-APL AML, standard for "fit" patients is intensive chemo ("7+3" regimen).
3. Genetics guide post-remission therapy. Favorable risk -> chemo only. Adverse risk -> Allogeneic Stem Cell Transplant.

MANAGING TREATMENT COMPLICATIONS

- Patient with APL on ATRA -> develops **fever, dyspnea, pulmonary infiltrates**: APL Differentiation Syndrome.
 - Management: Start Dexamethasone IMMEDIATELY. Do not stop ATRA unless symptoms are severe.
- Patient receiving consolidation chemo -> develops **ataxia, slurred speech**: High-Dose Cytarabine (HiDAC) Cerebellar Toxicity.
 - Management: STOP Cytarabine immediately. This is often irreversible.
- Patient develops decreased LVEF after chemotherapy: Suspect Anthracycline Cardiomyopathy.

Plasma Cell Dyscrasias (MGUS, Multiple Myeloma)

KEYWORDS / TYPICAL PRESENTATION

- Multiple Myeloma (MM): Bone pain, pathologic fractures, fatigue (anemia). Remember CRAB criteria for symptoms:
 - Calcium (>11), Renal insufficiency, Anemia, Bone lesions (lytic).
- MGUS: Asymptomatic patient with an incidentally found M-spike.